

JEE MAIN REVISION MASTER: CIRCLE SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 183 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Circle Equation, Tangents & Chord of Contact	178 PYQs	Circles
High	Tangents, Normals & Orthogonal Circles	10 PYQs	Tangents & Circles

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Circle Equation, Tangents & Chord of Contact	$r = \sqrt{g^2 + f^2 - c}$, $\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$, $\text{Chord of contact: } T = 0$, $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$
Tangents, Normals & Orthogonal Circles	$\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$, $\text{Length of tangent: } L = \sqrt{S_1}$, $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circle Equation, Tangents & Chord of Contact (Very High Yield — 178 PYQs)

Core Principle: Standard and general equations of circles, condition for line $y = mx + c$ to be tangent, and director circle $x^2 + y^2 = 2a^2$.

■ **Caution / JEE Pitfall:** The center of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(-g, -f)$, NOT $(+g, +f)$.

■ **Speed Trick:** The locus of the point of intersection of two perpendicular tangents to a circle is its director circle (radius $\sqrt{2}a$).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let C be a circle having centre in the first quadrant and touching the x -axis at a distance of 3 units from the origin. If the circle C ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x + (k-1)y + 3 = 0$ and $2x + \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the set of all values of r , for which the circles $(x+1)^2 + (y+4)^2 = r^2$ and $x^2 + y^2 - 4x - 2y - 4 = 0$ intersect at two distinct p ..." [Ans: N/A]

3.2 Tangents, Normals & Orthogonal Circles (High Yield — 10 PYQs)

Core Principle: Point and slope forms of circle tangents, director circle $x^2 + y^2 = 2a^2$, and orthogonality condition.

■ **Caution / JEE Pitfall:** The normal to a circle at any point ALWAYS passes through the center of the circle $(-g, -f)$.

■ **Speed Trick:** Common chord of two intersecting circles $S_1 = 0$ and $S_2 = 0$ is given directly by the linear equation $S_1 - S_2 = 0$.

Representative Source PYQs:

- **[2024 Morning Shift] Q#1 (MCQ):** "Let $C: x^2 + y^2 = 4$ and $C': x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect is (a, b) , then $a + b =$ _____" **[Ans: N/A]**
- **[2024 Evening Shift] Q#3 (MCQ):** "Let the circle $C_1: x^2 + y^2 - 2(x + y) + 1 = 0$ and C_2 be a circle having centre at $(-1, 0)$ and radius 2. If the line of the common chord of C_1 and C_2 is $x + y = k$, then $k =$ _____" **[Ans: N/A]**
- **[2024 Evening Shift] Q#3 (Numerical):** "Consider two circles $C_1: x^2 + y^2 = 25$ and $C_2: (x - \alpha)^2 + y^2 = 16$, where $\alpha \in (5, 9)$. Let the angle between the two radii at the point of intersection be θ . Then, the value of $\cos \theta$ is _____" **[Ans: 1575]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x + y = 1$ is $\sqrt{14}$, then the square of the radius of C is _____.

Correct Answer: 8

Solution Analysis: $r^2 = \left(\frac{\sqrt{14}}{2}\right)^2 + \left(r - \frac{1}{\sqrt{2}}\right)^2$ $r^2 = \left(\frac{7}{2}\right) + r^2 - \sqrt{2}r + \left(\frac{1}{2}\right)$ $\sqrt{2}r = 4$ $r = 2\sqrt{2}$ $r^2 = 8$